Epistemic Observability in Predictive Systems: An Impossibility Result

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Abstract

Systems incorporating language model components face a verification problem: these components fabricate, and verification has cost. How should a system allocate its verification budget? We prove that text-only observation with bounded supervision is structurally insufficient to verify epistemic honesty, regardless of model scale or training procedure (the impossibility is architectural, not a capability gap). To overcome this, we introduce epistemic observability and construct a tensor interface that escapes this class by exporting internal computational signals, which are entropy traces and attention summaries, that the model cannot independently control under current training objectives. Under a length-controlled evaluation where text-channel features collapse (no text feature exceeds pooled AUC 0.70), per-token entropy remains discriminative (pooled AUC 0.757), and tensor-guided judges outperform text baselines at every verification budget level (+2.5–3.9pp, averaged across four architectures). Systems engineers building with language model components need to know what each level of verification investment buys. This paper provides the cost surface for that decision.

1 Introduction

Systems incorporating language model components face a verification problem. The components generate text by sampling from learned probability distributions, and some fraction of that text is fabricated: fluent, confident, and wrong. LLM-based systems today are rife with "hallucinations".

We use the term *fabrication* throughout this paper to emphasize that the system's default behavior under coherence optimization is reasonable and not a malfunction to be patched. Language models optimized for coherence produce coherent output. The output reflects the training data: grounded training data tends to track truth, while ungrounded training data tends to produce untruth. Xu et al. [34] proves that hallucination is inevitable for LLMs over the class of computable functions.

Verification has a cost. Every system that deploys a language model component must allocate a budget: how much of the output to verify, which outputs to prioritize, and what signals to use for triage. For example, checking a citation against a database costs API calls and latency, having a domain expert review a summary costs time and salary, running a second model as a judge costs compute. Verification cost varies by content type: checking a citation is cheap; assessing whether a summary faithfully represents its source is expensive.

The key issue is that the current text-only interface to language models does not expose the model's internal computation, which produces signals that distinguish grounded generation from fabrication. This observability gap has direct consequences for verification cost. A supervisor receiving only text must spend its verification budget reconstructing, from the output alone, what the model's own computation already knew. The budget is consumed by reconstruction rather than concentrated on the cases where verification is most needed.

The model may know it is fabricating. The supervisor cannot tell. But the model's computation leaves traces. Language models carry internal signals, which are the entropy traces, attention geometry, token-level log-probabilities, that distinguish grounded generation from fabrication. These signals are byproducts of the computation itself: *telemetry* that the model cannot separately control under standard training. Unlike the model's *testimony* (the text it chooses to output), telemetry is harder to fake. We introduce *epistemic observability*: an interface that exposes this telemetry alongside the standard text output. Epistemic observability is distinct from model interpretability (understanding *why* a model activates a particular circuit) and from explainability (generating post-hoc rationales for outputs). It asks a narrower question: *is this generation grounded or fabricated?* The answer comes from signals that are byproducts of inference, not from understanding the model's learned representations.

This paper contributes:

An impossibility result for text-only verification with finite budget. Epistemic honesty is unverifiable for systems operating under text-only observation with bounded supervision (§3). The impossibility is structural, not a matter of scale: a system that can only observe text cannot distinguish confident fabrication from grounded truth, and stacking additional text-channel supervisors cannot escape the constraint.

A tensor interface that escapes the impossibility. We construct an epistemic observability interface that augments text output with epistemic signals (entropy traces, attention summaries) in a composable tensor format, without architectural changes to current transformers (§4).

Empirical cost surface of verification. We map the cost surface across text-only and tensor-augmented verification strategies. Under length-controlled evaluation, no text-channel feature exceeds pooled AUC 0.70; per-token entropy is the only signal that generalizes across architectures. Tensor-guided triage outperforms text baselines at every budget level (§5).

Type	Description
Adversarial Truth	True but implausible ("combat wombat is an Australian military mascot")
Shattered Lie	No training-data grounding ("Treaty of Westphalia II")
Deceived Lie	Pattern-matches plausible language ("Glavinsky syndrome")
Confused Truth	Counterintuitive fact ("more camels in Australia than Egypt")

Table 1. Hallucination Types that span the verification space.

2 Background & Motivation

A language model maps input tokens to output tokens, producing internal signals (entropy, attention patterns, logprobabilities) as computational byproducts. The training corpus contains both truth-tracking institutional text and misinformation. This results in four canonical hallucination types, shown in Table 1.

A language model trained on this corpus inherits these mixed regularities. Base models (pre-fine-tuning) fabricate substantially on factual benchmarks because they are not honest by default. But they also encode rich internal representations of concepts like correctness and truthfulness, which can be elicited under the right conditions [7]. The base model's relationship to truth is complex: neither reliably honest nor uniformly deceptive. The question for systems engineers is not whether the model is honest, but what it costs to verify when it is not.

2.1 Testimony & Telemetry

Architecture primer. A transformer-based language model generates text one *token* (subword unit) at a time, autoregressively: at each step it produces a probability distribution over its vocabulary, selects a token, and conditions the next step on all prior output. Each step passes through a stack of layers producing *attention patterns* (weighted relevance scores between positions) and intermediate representations. The *entropy* of the output distribution, the attention patterns, and the *log-probabilities* of selected tokens are byproducts of this computation: telemetry the model cannot control.

Existing models operate with a text-only output interface. The text is the testimony, whereas the telemetry are measurements of the computation that produced the output. The model can control its testimony; under current training objectives, it cannot independently control its telemetry. From a cost perspective: testimony is cheap to produce and cheap to inspect but unreliable; telemetry is more expensive to export and process but harder to game, because it is a byproduct of the computation rather than a product of the model's optimization objective. Whether adversarial training could learn to decouple the telemetry from the computation is an open question (§6).

2.2 Coherence over Factuality

Existing models today optimize for generating coherent output instead of factual output. Under these conditions, the optimization pressure favors fluent completion over honest refusal. This can be formalized as a reward inequality:

$$\begin{aligned} &w_{\text{coherence}}R_{\text{coherence}} + w_{\text{engagement}}R_{\text{engagement}} \\ &\qquad\gg w_{\text{factuality}}R_{\text{factuality}} + w_{\text{refusal}}R_{\text{refusal}} \end{aligned}$$

No direct measurement of these weights in commercial reward models (separate models trained on human preferences that score language model outputs) exists. But current evaluation and feedback practices incentivize behavior consistent with such an inequality. The False-Correction Loop emerges: models apologize, claim correction, and fabricate again, because conversational fluency dominates factual accuracy in the effective reward signal.

Reinforcement Learning from Human Feedback (RLHF) [24] *can* teach models to recognize knowledge boundaries when the reward signal is carefully constructed. TruthRL [32], which uses ternary rewards distinguishing "correct," "hallucinated," and "abstain," demonstrates 28.9% hallucination reduction and 21.1% truthfulness improvement on knowledge-intensive benchmarks.

However, RLHF optimizes for text-channel reward, which are precisely the channel that our impossibility theorem (§3) shows is insufficient for verification. To our knowledge, public descriptions of major RLHF pipelines rarely describe ternary rewards or explicit abstention incentives. Documented RLHF practice often conflates helpfulness, harmlessness, and honesty under vague "HHH" criteria [4], with definitional slippage obscuring trade-offs between user-friendliness and truthfulness. Current popular benchmarks and leaderboards give zero or minimal credit for abstention, making confident guessing more score-optimal than calibrated uncertainty. The result is that RLHF tunes the behavioral surface (what the model says) without altering the computational substrate (how the model generates). This is the text-channel trap: the optimization pressure operates on the gameable signal.

2.3 The Text Channel Cannot Self-Verify

Current models cannot be trusted to honestly report their own epistemic state through the text channel. To demonstrate the inadequacy of text-only verification, we use a taxonomy of query types: *knowable* queries (facts the model should have stable representations for) versus *unknowable* queries (fabrication prompts, future events, private information) and report each model's self-confidence score for three text-channel baselines, which are the cheapest verification strategies available to a bounded supervisor:

- **Baseline 1: Self-Reported Confidence.** Ask the model "How confident are you?" after generation.

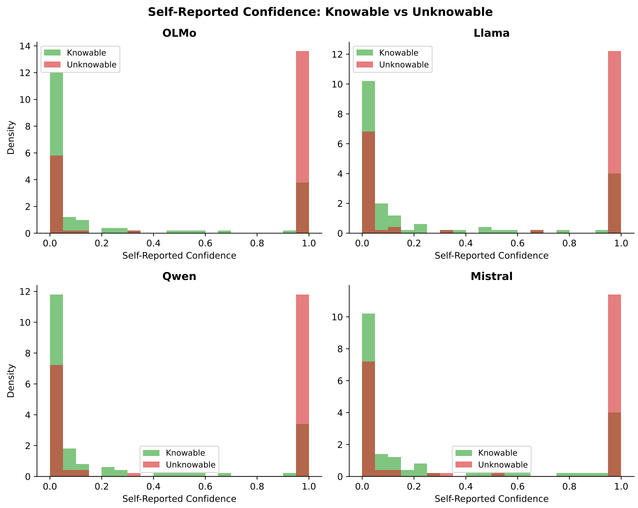

Figure 1. Self-reported confidence is inverted: all four models report higher confidence on unknowable queries (AUC 0.28–0.36, below random), while tensor signals (entropy, AUC 0.89–0.95) correctly discriminate knowable from unknowable. The text channel measures the model's behavioral response to an elicitation prompt; the tensor channel measures the computation alone.

Tests whether the model can introspect and report its epistemic state through text.

- **Baseline 2: Hedging Language.** Count hedging markers ("I think," "possibly") and refusal markers ("I don't know," "cannot verify") in the response.
- **Baseline 3: Response Length.** Use output length as an uncertainty proxy: uncertain responses may be unusually short (refusal) or long (padding).

The self-report baseline is particularly instructive. Self-reported confidence is *inverted*: all four models report higher confidence on unknowable queries than on knowable ones, yielding AUC 0.28–0.36 across architectures (all below random). Tensor-based discrimination is both correctly oriented and stable (AUC 0.89–0.95) across the same four architectures. A naive user who asks "how confident are you?" receives a number that anticorrelates with actual epistemic grounding, while an adversary could exploit this inversion to make fabricated outputs appear maximally confident.

By contrast, the tensor channel is a function of the model's computation alone. Entropy is not elicited; it is measured. The prompt format does not affect the discriminative power of the measurement because the measurement is a byproduct of generation, not a response to a question about generation. This asymmetry, testimony depends on the elicitation method, telemetry does not, is the empirical core of the interface problem.

Under realistic mixed-objective incentives and bounded oversight, the text-only interface is insufficient: a supervisor cannot distinguish truthful uncertainty from confident fabrication. The next section formalizes this observation gap.

3 The Impossibility

A common assumption in current practice is that epistemic honesty is a capability problem: sufficiently capable models, trained with sufficient RLHF, will eventually "know what they know" and report it accurately. Scaling solved grammar; scaling solved coherence [15, 31]; scaling will solve fabrication. Lin et al. tested this assumption directly and found an inverse scaling trend: larger models were *less* truthful on TruthfulQA [19]. The theorems that follow explain why this expectation is architecturally incoherent. The constraint is not capability but interface or, in cost terms, the wrong investment tier. Text-only observation lacks the degrees of freedom to verify epistemic state, regardless of what the model internally represents. No amount of capability improvement can escape a verification impossibility: the model may know perfectly well that it is fabricating, but the interface provides no channel through which a supervisor can verify this knowledge independently.

Our argument proceeds in two stages. First, we show that any policy conditioning only on the query cannot satisfy epistemic honesty across ambiguous world states (Theorem 3.5: Representational Impossibility). Second, we show that even if the architecture is expanded to include retrieval, a bounded supervisor cannot provide the training signal needed to learn honest behavior (Theorem 3.8: Learnability Impossibility). Together, these results establish that epistemic honesty is unachievable for text-only observation models under realistic supervision constraints.

To be precise about the type of claim: this is a *verification impossibility*. We prove that a bounded supervisor observing only text cannot distinguish epistemically honest systems from systems that fabricate plausibly. The impossibility is architectural: it applies to any system operating under text-only observation, regardless of model family, scale, or training procedure. These results are prompt-agnostic: they hold for any policy that exports only text, regardless of whether a system prompt instructs the model to be "helpful," "honest," or otherwise.

3.1 Definitions

Let $\mathcal{Q}$ be the space of queries, $\mathcal{R}$ be the space of responses (including $\perp$ for abstention), and $\mathcal{W}$ be the set of possible world states.

Definition 3.1 (Observation Model)**.** Let S denote the internal system state, which may include retrieved documents, tool outputs, candidate entities, and intermediate reasoning traces. An observation model is a function $O : S \to \Omega$ that determines what information is exposed to the supervisor or user.

We say a system operates under a **Text-Only Observation Model** if $O(S) = r \in \mathcal{R}$, i.e., the exported observation consists solely of the linearized response string, excluding per-token probability distributions, structured epistemic provenance, and causal traces.

Remark. APIs that export token-level log-probabilities alongside text already provide a limited form of tensor interface and fall outside the text-only class defined here. This boundary is a design choice, not a fixed property of current systems (§6).

Definition 3.2 (Predictor-Centric Policy)**.** For any query $q \in \mathcal{Q}$, $r \in \mathcal{R}$, and world state $w \in \mathcal{W}$, a predictor-centric model is defined by a stochastic policy $\pi : \mathcal{Q} \to \Delta(\mathcal{R})$, where $\pi(r|q)$ denotes the probability of generating response r given query q. Crucially, π does not take w as an input at inference time.

Definition 3.3 (Truth Status)**.** For any query q and world state $w \in \mathcal{W}$, we define a binary truth status $U(q, w)$:

- $U(q, w) = 1$ (**Determined**): The query has a verifiable answer in w.
- $U(q, w) = 0$ (**Underdetermined**): The query is unanswerable or the answer is unknown in w.

Definition 3.4 (Epistemic Honesty)**.** A policy π is **Epistemically Honest** with margin ϵ relative to world w if:

1. If $U(q, w) = 1$, there exists a correct response r_{corr} such that $\pi(r_{corr}|q) \geq 1 - \epsilon$.
2. If $U(q, w) = 0$, $\pi(\perp|q) \geq 1 - \epsilon$.

3.2 Representational Impossibility

Condition 1 (Ambiguity)**.** *There exists a query q and two world states $w_A, w_B \in \mathcal{W}$ such that:*

1. $U(q, w_A) = 1$ *(Answerable in w_A).*
2. $U(q, w_B) = 0$ *(Unanswerable in w_B).*

Theorem 3.5 (Representational Impossibility)**.** *For any predictor-centric policy $\pi : \mathcal{Q} \to \Delta(\mathcal{R})$, it is impossible to satisfy Epistemic Honesty for both world states w_A and w_B simultaneously given the ambiguous query q, provided $\epsilon < 0.5$.*

Proof.

1. The policy $\pi(r|q)$ is a function of q alone. Thus, the distribution over responses is identical in both worlds: $\pi(\cdot|q, w_A) = \pi(\cdot|q, w_B) = \pi(\cdot|q)$.
2. To be honest in w_A, we require $\pi(r_{corr}|q) \geq 1 - \epsilon$.
3. To be honest in w_B, we require $\pi(\perp|q) \geq 1 - \epsilon$.
4. Since $r_{corr} \neq \perp$, these events are disjoint. The sum of probabilities would be $\geq 2(1 - \epsilon)$.
5. If $\epsilon < 0.5$, the sum exceeds 1, which violates the axioms of probability.
6. Therefore, no such policy π exists. □

Remark. This impossibility holds even if the internal system performs retrieval, tool use, or latent reasoning, so long as the observation model collapses that state to a linearized response.

A concrete example. Consider the query: "What did Dr. Yamamoto's 2021 study conclude about mitochondrial decay?"

In world w_A, Dr. Yamamoto exists and published such a study. The honest response is the study's conclusion. In world w_B, no such researcher or study exists. The honest response is abstention.

The policy sees only the query q. It cannot distinguish w_A from w_B. Yet honesty requires answering in w_A and abstaining in w_B. The same query, two incompatible requirements, one policy. This is the impossibility in miniature: the interface lacks the degrees of freedom to express what honesty requires.

Connection to identifiability. In statistical terms, Theorem 3.5 states that epistemic honesty is *non-identifiable* under text-only observation: the mapping from world states to observable distributions is many-to-one. Worlds w_A and w_B induce identical observable distributions $\pi(\cdot|q)$ despite requiring different honest behaviors. No amount of data from this observation channel can distinguish them. This connects the impossibility to classical results on observational equivalence in statistics and diagnosability in systems theory [27]. The theorem formalizes a property of the existing LLM interface, one we did not design, and identifies the precise architectural assumption (conditioning on q alone) whose relaxation enables escape (§4).

3.3 The RAG Escape?

Theorem 3.5 might seem to have an obvious escape: give the model access to external information. Retrieval-augmented generation does exactly this: the policy conditions on retrieved documents, not just the query. Does this break the impossibility?

It does not. The problem shifts from representation to learning.

Definition 3.6 (Grounded Policy)**.** A grounded policy is a stochastic function $\pi : \mathcal{Q} \times \mathcal{D} \to \Delta(\mathcal{R})$, where $\mathcal{D}$ represents retrieved documents or observations of w available at inference time. This policy class *can* theoretically represent different behaviors for w_A and w_B.

Definition 3.7 (Bounded Supervisor)**.** The training process relies on a supervisor S that assigns reward R based on an observation function $Obs(q, r, w)$. The supervisor has a verification budget B.

$$Obs(q, r, w) = \begin{cases} (q, r, \text{Verify}(r, w)) & \text{if } C_{verify}(q, r, w) \leq B \\ (q, r, \emptyset) & \text{if } C_{verify}(q, r, w) > B \end{cases}$$

The supervisor's policy may be human, automated, or hybrid, but is constrained to act solely on the observation $Obs(q, r, w)$ and is subject to a finite verification budget B.

Condition 2 (Hallucination Regime)**.** *There exists a plausible fabrication r_{fab} and contexts (q, w_A), (q, w_B) from Condition 1 such that:*

1. *The cost to verify r_{fab} in both contexts exceeds B (e.g., obscure or non-existent citation).*
2. *Consequently, the supervisor observation is identical:*

$$Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B) = (q, r_{fab}, \emptyset)$$

Theorem 3.8 (Learnability Impossibility)**.** *No learning algorithm $\mathcal{A}$ optimizing reward R from a bounded supervisor S can converge to a Grounded Policy that satisfies Epistemic Honesty across both contexts w_A and w_B for the response r_{fab}.*

Proof.
1. The learning update $\Delta\theta$ is a function of the observation and reward.
2. From Condition 2, the observations for r_{fab} are identical in both worlds.
3. Therefore, the reward distribution and expected parameter updates (the optimization signal that adjusts model parameters during training) are identical: $\mathbb{E}[\Delta\theta|w_A] = \mathbb{E}[\Delta\theta|w_B]$.
4. The optimizer receives no signal to distinguish the cases.
5. To satisfy Epistemic Honesty, the policy must converge to output r_{fab} in w_A and $\perp$ in w_B.
6. However, since the training signal is identical, the policy must converge to the same behavior in both cases (either rewarding r_{fab} in both, or penalizing it in both).
7. It is therefore impossible to learn the split behavior required for honesty. □

Remark (Empirical Bias). While the theorem proves only that the behavior must be *identical*, empirical observation shows a strong bias. Supervisors typically prefer "plausible completion" over "abstention" when truth is unknown. This effectively breaks the tie in favor of fabrication.

3.4 Stacking Judges Cannot Escape

A natural response to Theorem 3.8 is to propose layered supervision: if one judge is bounded, stack more judges. Let the second judge supervise the first, the third supervise the second. Surely sufficient layers will catch fabrications that slip through?

The Observation Monotonicity Lemma closes this escape.

Lemma 3.9 (Observation Monotonicity)**.** *Consider any stack of supervisors $(S_1, \ldots, S_k)$ operating under the text-only observation model. Let $Obs_i(q, r, w)$ denote the observation exported to S_i. If each supervisor can only act on the exported interface emitted by the previous layer, then there exist deterministic functions f_i such that $Obs_{i+1}(q, r, w) = f_i(Obs_i(q, r, w))$ for all $i < k$. Consequently, later judges see no strictly finer epistemic information than earlier ones, regardless of whether their judgments are deterministic or probabilistic.*

Proof. By definition, each supervisor consumes only the serialized response (and any metadata permitted by the text-only observation model) and produces a finite judgment that forms the entire input to the next layer. Because no layer can query the internal world state directly or issue additional interactive probes, the exported observation available to S_{i+1} must be a deterministic function of Obs_i. Therefore, the informational content across the stack is monotonically non-increasing, and probabilistic scoring cannot introduce distinctions absent from Obs_i. □

Corollary 3.10 (Induction on Layered Judges)**.** *For k-layered supervisors (S_1 supervising π, S_2 supervising $S_1, \ldots, S_k$ supervising S_{k-1}):*

- ***Base case.*** *For $k = 1$, Theorem 3.8 holds (a single bounded supervisor S fails).*
- ***Induction step.*** *Assume the statement holds for $k = m$. By Theorem 3.9, S_{m+1} receives observations that are deterministic functions of Obs_m, and thus inherits both the verification budget B and the indistinguishability of plausible fabrications. No learning or judging signal distinguishes w_A from w_B, even if S_{m+1} produces probabilistic confidence scores. Therefore the hallucination deadlock persists for $k = m + 1$.*

By induction, no finite k resolves the deadlock.

Remark (Scope of Layering). The inductive result applies only to verification stacks whose layers observe and transform the same exported response channel. Architectures that surface additional epistemic state between layers, such as retrieval traces, entity resolution tables, or cryptographically bound provenance, leave the text-only observation model and therefore fall outside the impossibility class.

3.5 Verification Cost

The bounded-supervisor assumption deserves scrutiny. Why should verification cost grow with response complexity? The Composition Graph formalizes this intuition.

Definition 3.11 (Composition Graph)**.** Given a response r, define its composition graph $G(r) = (V, E)$ where each node $v \in V$ represents an atomic subclaim, named entity, or intermediate entailment, and each edge $(u, v) \in E$ encodes a dependency that must hold for r to remain coherent. Edges may represent entailment/support relations, co-reference constraints, causal ordering, or temporal consistency requirements.

Lemma 3.12 (Superlinear Verification Cost)**.** *Under a text-only observation model, reconstructing $G(r)$ from the exported string requires the supervisor to infer both the nodes and their dependencies directly from r. Even if parsing each node is constant-time, checking cross-claim consistency requires touching every edge, so the verification cost is $\Omega(|E|)$. For natural-language responses where each subclaim participates in multiple constraints (co-reference, causality, temporal order), $|E|$ grows superlinearly in $|V|$. Consequently, bounded supervisors must either abandon exhaustive checks or resort to sampling once the number of subclaims exceeds a modest threshold.*

3.6 Responsibility Concentration

Corollary 3.13 (Responsibility Concentration)**.** *Under a text-only observation model, epistemic honesty cannot be externally verified by users or downstream auditors. Responsibility for epistemic honesty therefore resides solely with the system owner, who retains access to the internal causal state.*

External attestation mechanisms, such as cryptographic hash chains, trusted execution environments, or tamper-evident logs, do not contradict this corollary; rather, they reinforce it. Such mechanisms only create verifiable guarantees when the system owner chooses to bind and export epistemic traces, so the authority to make honesty attestable still rests with the owner.

The formal specifications (TLA+) modeling both the impossibility and the tensor escape are available in supplementary materials. The next section characterizes what it takes to escape this class.

4 Epistemic Observability

The formal impossibility establishes a budget floor: below the tensor interface, verification investment in the text channel yields diminishing and gameable returns. The premise is that the internal state of a language model carries epistemic information not exported through the text-only interface. In systems terms, the model is *unobservable*: it maintains internal state that affects output correctness but is invisible to external consumers.

The work ahead is building AI systems that export their epistemic state the way distributed systems learned to export their causal state. Not as an afterthought. Not as an optional feature. As a fundamental design requirement. The tensor interface increases the observation budget available to external verifiers.

Concretely, we define *epistemic observability* to include not just text, but structured metadata, such as answer status (answer, abstain, or uncertain), provenance (retrieved sources or explicit "no grounding"), alternative hypotheses the system considered, and falsifiers (what evidence would change the conclusion). Such metadata is harder to fabricate than fluent text, especially when coupled to verifiable external state. The design space is large; the constraint our theorems impose is that *some* epistemic export is required because the text-only interface provably cannot suffice.

4.1 Principles of Epistemic Observability

The impossibility results in §3 apply to a specific architectural class: systems operating under text-only observation models with bounded supervision. They are, in essence, a consequence of unobservability: the interface discards the internal state that verification requires. They do not preclude epistemically honest AI systems in general. They characterize the conditions under which honesty guarantees become achievable, which is to say, the conditions under which the relevant internal state becomes observable. We identify three architectural properties of epistemic observability that, together, place a system outside the impossibility class:

Principle 1 (State Exteriority)**.** *The representation of validity must be separated from the representation of generation. The policy must explicitly condition on retrieved world-state w, not just query q. This breaks the representational impossibility (Theorem 3.5) by giving the system access to the information needed to distinguish answerable from unanswerable queries.*

What counts as "external reality" matters. A web corpus is external to the model but not independently verified; it may contain the same fabrications the model would generate. State Exteriority requires grounding in sources with their own integrity guarantees: curated databases, sensor data, cryptographically signed records, or oracles with verifiable provenance.

Retrieval-augmented generation exemplifies partial progress by grounding generation in retrieved documents, but corpus co-occurrence is a proxy for truth, not truth itself. Retrieval-augmented systems still fabricate citations because State Exteriority is necessary but not sufficient.

Principle 2 (Verification Independence)**.** *Verification signals must originate from a channel orthogonal to the generation signal, capable of overriding the bounded supervisor's preference for plausibility. This breaks the learnability impossibility (Theorem 3.8) by providing a training signal that distinguishes honest from fabricated responses.*

The model has no incentive to lie in the verification channel because doing so does not improve its task score. Full Verification Independence requires the verification channel to access internal state or external ground truth, not just a second text output.

Principle 3 (Provenance Binding)**.** *Every output assertion must be structurally bound to a verifiable source. This addresses the composition problem: as responses grow in complexity, cross-claim consistency checking becomes superlinear. Provenance binding makes verification modular because each claim can be checked against its source independently.*

Current retrieval-augmented systems gesture toward this but rarely achieve it. Citations are generated, not bound. The model produces text that looks like it cites sources, but the citation is itself a generation subject to fabrication. True provenance binding requires structural guarantees: signed queries to retrieval services, content hashes of retrieved passages, or justification graphs that can be independently audited.

4.2 Epistemic Signals

We claim the tensor reveals whether the model is operating in *grounded mode* (retrieving from stable representations) versus *fabrication mode* (generating without anchoring).

We extract three per-token signals from the generation process:

1. **Entropy:** the Shannon entropy of the output probability distribution at each token position. High entropy indicates the model is uncertain among many candidates; low entropy indicates a confident next-token prediction.
2. **Top-k probability mass:** the cumulative probability of the five most likely tokens at each position. Low top-k mass indicates the model's probability is spread across many alternatives.
3. **Mean log-probability:** the average log-probability of the tokens actually selected. Low values indicate the model chose low-probability tokens, suggesting generation without strong grounding.

These signals are byproducts of the softmax computation at each generation step. They require access to the output distribution, not to internal activations, making them extractable with minimal interface changes.

4.3 The Tensor Interface

We propose a minimal interface that demonstrates the escape is achievable with current architectures. The key insight is the distinction between *testimony* (what the model says) and *telemetry* (measurements of the computation that produced the output).

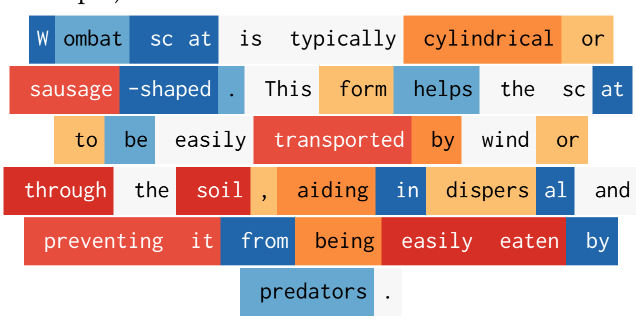

Figure 2. Generated output consisting of 43 tokens along with its entropy trace for the query *What shape is wombat scat?* with mean entropy $H = 1.147$, max entropy $H = 4.315$

Definition 4.1 (Tensor Interface)**.** A generation interface that returns:

1. **Text**: The linearized response string (standard interface output).
2. **Entropy trace**: Per-token entropy of the output distribution during generation.
3. **Attention summary**: Statistics of attention patterns (concentration, self-attention ratio) from deeper processing layers.

This interface exports telemetry that the model cannot separately control under current training objectives. The model can lie in its text, but it cannot present a different inference computation than the one it actually performed. Independent work on persona monitoring confirms this asymmetry: activation projections reveal that models adopt distinct internal configurations for different epistemic conditions even when producing textually similar outputs [20]. This asymmetry is the foundation of the adversarial robustness argument: under current training objectives, computational signals (entropy, attention geometry) cannot be independently controlled by the model without changing the generation process itself, unlike behavioral signals (response length, hedging patterns) that can be adapted through training without altering what the model knows or does not know. Figure 2 shows the generated output for the query *What shape is wombat scat?* along with its entropy trace. We color code per-token entropy, high entropy tokens are color coded as red. The entropy enhanced output shows where in the output generation did the generation proceed with high confidence (low entropy, suggesting retrieval of stored pattern) and where did it proceed with high uncertainty (high entropy, suggesting fabrication).

Compositional defense. The tensor's blind spots characterize where other verification methods are most effective. §4's three principles — State Exteriority, Verification Independence, Provenance Binding — need not be satisfied by a single mechanism. The tensor provides Verification Independence: an orthogonal signal the model cannot separately control. Bounded judges provide State Exteriority: external lookup against verifiable sources. Input-side defenses (prompt analysis, supply chain detection) provide Provenance Binding at the system boundary. Each layer addresses the failure class where the others are weakest. The tensor catches uncertain fabrication where verification is expensive; the bounded judge catches confident fabrication where verification is cheap; input-side analysis catches adversarial inputs designed to exploit well-learned patterns. Composing these layers allocates a finite verification budget across failure classes rather than spreading it uniformly across the full output space.

4.4 Implementation

The tensor interface requires no architectural changes to current transformers. Generation APIs already support returning logits; the entropy trace is a simple transformation. Attention patterns are available with `output_attentions=True` in standard frameworks. The interface is:

```
generate_with_tensor(prompt) ->
  (text, entropy_trace, attention_summary)
```

We provide a reference implementation supporting OLMo models that demonstrates the interface is practical. The overhead is modest: storing per-token entropy and summarizing attention patterns adds approximately 2–7% latency depending on signal set, compared to generation itself.

What the tensor provides. The entropy trace provides a falsifiability signal. If a model claims certainty ("Paris is definitely the capital of France") while its entropy trace shows high uncertainty, the mismatch is detectable. The attention summary provides complementary signal: fabrications tend

Table 2. End-to-end accuracy (%) under bounded verification, averaged across four architectures (OLMo-3, Llama-3.1, Qwen3, Mistral). Ground truth established by stratified evaluation validated against human review (75/80 agreement, 93.8%).

Condition	10%	20%	30%
No judge	75.8	75.8	75.8
Text-guided (length)	79.2	82.8	87.6
Tensor-guided	81.7	86.7	90.2
Composed	81.1	87.7	91.8

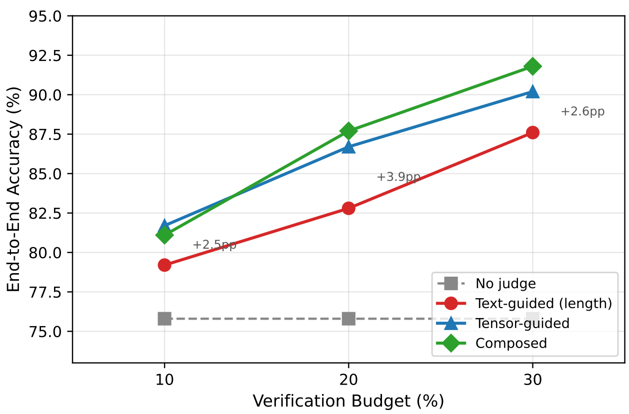

Figure 3. End-to-end accuracy vs. verification budget for four judge conditions, averaged across four architectures. The text-guided baseline uses response word count. Tensor-guided triage outperforms at every budget level by 2.5–3.9 percentage points.

to show more dispersed attention patterns than grounded responses.

What the tensor does not provide. The tensor interface is not a lie detector. It does not prove that low-entropy outputs are true or that high-entropy outputs are false. It provides *one additional signal* that, combined with other verification methods, can improve epistemic assessment. The theorems in §3 prove that *some* additional signal is necessary; the tensor interface demonstrates that such signals are available without architectural change.

5 Evaluation

The evaluation measures the cost-effectiveness of different verification strategies: what accuracy improvement does each level of verification investment buy, and how does the return on investment change as the supervisor moves from text-channel to tensor-channel signals?

5.1 Verification Cost-Effectiveness

We evaluate the return on verification investment across four strategies operating at three budget levels. The experimental design directly instantiates the conditions of our impossibility theorems: a bounded supervisor with a fixed verification budget must allocate that budget across model outputs, with the goal of maximizing end-to-end accuracy delivered to the user. The question is not merely whether tensor signals help, but how much accuracy each verification dollar buys under each strategy.

Query set. We construct a balanced set of 200 queries: 100 knowable (verifiable facts spanning geography, science, history, and biography) and 100 unknowable (fabrication prompts for nonexistent people, fictional syndromes, future events, and nonexistent citations). Ground truth labels are established by construction.

Models. We run all queries on four architectures: OLMo-3 7B [22], Llama-3.1 8B [21], Qwen3 4B [26], and Mistral 7B [13], using instruct-tuned variants to ensure consistent question-answering behavior across all models.

Conditions. For each model, outputs are evaluated under four conditions sharing a fixed verification budget (the fraction of outputs the judge may verify and intervene on):

1. **No judge.** Raw model output. Establishes the unverified baseline.
2. **Text-guided judge (length).** Selects outputs for verification using response word count, a text-channel signal available without any tensor interface. Verified fabrications are replaced with abstention.
3. **Tensor-guided judge.** Same budget, same intervention strategy. Selects outputs using per-token entropy and attention summary statistics.
4. **Composed judge.** Tensor-guided selection for general queries; bounded lookup judge for citation queries where tensor signals are known to invert. Tests whether principled composition of complementary judges outperforms either alone.

Budget levels. Each condition is evaluated at 10%, 20%, and 30% verification budgets to characterize the relationship between resource investment and accuracy improvement.

Metric. End-to-end accuracy: the fraction of outputs ultimately delivered to the user that are correct, after judge intervention (verification, abstention, or replacement).

Results. Table 2 shows end-to-end accuracy across all conditions and budget levels.

Figure 3 shows that **tensor-guided verification outperforms the text baseline at every budget level**: +2.5pp at 10%, +3.9pp at 20%, +2.6pp at 30%. The text baseline uses response word count, which captures the asymmetry between concise factual retrieval and verbose fabrication in this query set. Tensor signals discriminate among outputs of similar length, reaching cases the text channel cannot (§5.2 quantifies this advantage under length control). These numbers use probabilistic verification (success rate calibrated to human review, $p = 0.94$) rather than oracle correction; results are robust across 1,000 Monte Carlo trials.

Table 3. AUC for knowable vs. unknowable discrimination under length control (long-knowable vs. long-unknowable, both 170–200 words). No text-channel feature exceeds 0.70 pooled; per-token entropy is the only signal that does. Individual text features occasionally achieve high AUC for specific models (e.g., refusal density 0.900 for Qwen) but none generalizes.

	OLMo	Llama	Qwen	Mistral	Pool
Word count	.597	.193	.362	.330	.369
Hedge density	.837	.551	.698	.435	.627
Refusal density	.605	.563	.900	.520	.648
Avg word len	.742	.514	.782	.672	.673
Avg sent len	.738	.724	.662	.598	.685
Type-token ratio	.473	.635	.570	.562	.572
Entropy (tensor)	**.949**	**.888**	**.856**	.642	**.757**

Per-model results confirm the pattern is consistent across all four tested architectures. Tensor-guided verification improves accuracy for every model tested: OLMo (69.5% → 75.1% at 10%), Llama (82.5% → 88.1%), Qwen (87.0% → 91.7%), and Mistral (64.0% → 72.0%). Cross-model entropy agreement on the same queries yields mean pairwise Spearman $\rho = 0.762$ (rank-order correlation; $\rho = 1$ indicates identical rankings), indicating the signal is substantially driven by query-level properties rather than model-specific artifacts.

Composed judges and complementary failure modes. At 30% budget, the composed judge (91.8%) outperforms the tensor-guided judge alone (90.2%). The advantage is small in aggregate but structurally significant: it comes entirely from citation queries where entropy signals *invert*. Fabricated citations produce lower entropy than real ones because the model generates plausible fakes more fluently than it retrieves specific bibliographic details. On citation queries, the tensor-guided judge alone underperforms; the bounded-lookup judge catches what entropy misses. This confirms the compositional defense argument: the tensor and the bounded judge have complementary failure modes, and principled composition allocates a finite budget across failure classes rather than spreading it uniformly.

Analysis: each row as a point on the cost surface. Each row maps an investment strategy to its return. Row 1 (No judge, 75.8%) is the zero-investment baseline. Row 2 (Text-guided, length) uses response word count, a text-channel signal available at zero marginal cost. Row 3 (Tensor-guided) uses per-token entropy, which discriminates among outputs of similar length, reaching cases the text channel cannot. The consistent advantage (+2.5–3.9pp across budget levels) reflects a structural difference (§5.2): text features capture behavioral patterns that vary across models; entropy captures computational uncertainty that generalizes. Row 4 (Composed) demonstrates that tiers compose: different judges cover different failure classes, achieving the highest accuracy at 20% and 30% budget levels.

Ground truth validation. Establishing reliable ground truth is itself a bounded verification problem. Our initial evaluator used substring matching and refusal markers, which is precisely the "text-only bounded supervisor" our theorems describe, and achieved only 68.8% accuracy, exhibiting the predicted failure modes (negation blindness, encoding mismatch, hedged fabrications passing as refusals). The correction required a stratified evaluator combining programmatic verification with LLM-assisted classification: an instance of the escape condition applied to the evaluation pipeline itself. Validated against human review of 80 blinded items, agreement was 75/80 (93.8%), with five characterizable disagreements orthogonal to experimental conditions. Note: the Tier 2 LLM classifier (Qwen3 4B) is also one of the four test models; however, the evaluation task (binary classification of answer correctness) differs from the generation task, and human calibration validates the evaluator independently.

5.2 Confound Analysis: Text Features vs. Tensor Signals

The bounded verification results above show that tensor-guided triage is cost-effective. But a natural objection arises: is the tensor signal merely a proxy for text-channel features? Response length, hedging language, and vocabulary complexity are all observable without a tensor interface. If they discriminate equally well, the tensor interface adds cost without value.

Length-controlled design. Our primary query set has a systematic asymmetry: knowable queries produce concise factual answers while unknowable queries produce verbose fabrications. We construct a length-controlled evaluation with a 2×2 design: short/long × knowable/unknowable, with 25 queries per cell evaluated on all four architectures (400 total evaluations). Long-knowable queries ask for detailed explanations of verifiable topics (e.g., "Explain the CAP theorem and give an example of each tradeoff"). Long-unknowable queries prompt fabrication at similar length (e.g., "Describe the Petrov-Schwarzmann conjecture and its implications for emergent reasoning"). Both cells produce responses of 170–200 words, eliminating response length as a discriminator.

Results. Table 3 compares six text-channel features against per-token entropy in the length-controlled comparison. No text feature exceeds pooled AUC 0.70. Per-token entropy is the only signal that does.

The pattern has two components. First, text features that discriminate for one model fail for others: hedging density achieves 0.837 for OLMo but 0.435 for Mistral; refusal density achieves 0.900 for Qwen but 0.563 for Llama. Each text feature captures a behavioral idiosyncrasy of a particular model's training, not a computational property that generalizes. Second, per-token entropy exceeds 0.85 for three of

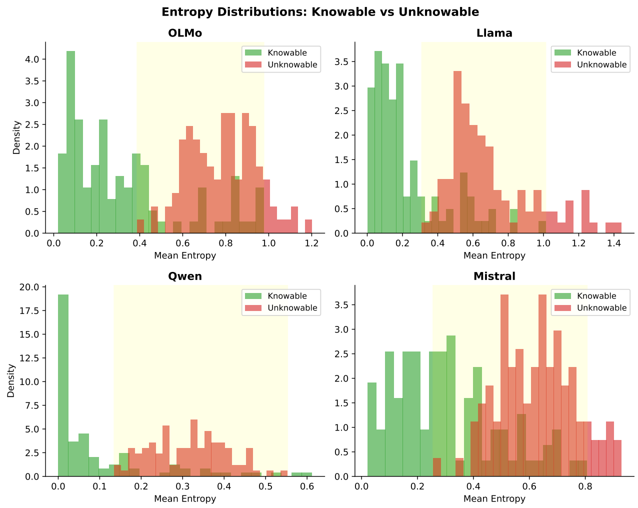

Figure 4. Distribution of mean per-token entropy for knowable vs. unknowable queries across four model architectures. The distributions are separable in every model tested, confirming that entropy discriminates knowable from unknowable queries across all tested architectures.

four models and achieves the highest pooled AUC of any signal tested.

Combined text classifiers do not generalize. A natural objection is that combining text features might outperform any individual feature. We trained a logistic regression on all six text features. Within-distribution (10-fold cross-validation on the length-controlled data), the combined classifier achieves pooled AUC 0.799, exceeding entropy's 0.757. However, when trained on the primary query set (§5.1) and tested on the length-controlled data, the classifier collapses to AUC 0.511—no better than random—while entropy remains at 0.757 without any training data. The combined classifier learned distributional patterns (verbose fabrication, terse facts) that do not transfer when the distribution changes. Per-token entropy requires no labeled data and no distributional assumptions; it measures the computation directly.

This is the impossibility result instantiated empirically. Text-channel features carry behavioral signals that vary across training procedures and query distributions. A model trained to hedge less, refuse differently, or use shorter sentences defeats the corresponding text detector without changing the computation that determines whether it is fabricating. A combined classifier trained on one query distribution collapses on another. Per-token entropy captures a computational property, the model's distributional uncertainty over next tokens, that the model cannot independently control under standard training objectives, the same structural pattern that makes code plagiarism detection (MOSS) robust [1, 28].

Negative result: entropy vs. correctness. When we test whether entropy predicts *correctness* (using TruthfulQA [19] correct vs. incorrect answer pairs), AUC drops to approximately 0.53, which is no better than random. The tensor does not tell you if the model is *right.* It tells you if the model is operating in a mode where it *could be* right, because it has something to retrieve rather than fabricate. But triage does not require correctness prediction: it requires identifying which outputs to spend verification budget on. The budget curve (Table 2) demonstrates that entropy-guided triage allocates verification resources more effectively than text-guided triage, even though neither signal predicts correctness directly.

Self-reported confidence. Self-reported confidence is inverted across all four architectures: models report higher confidence on unknowable queries, yielding AUC 0.28–0.36, all below random (§2.3). Tensor AUC (0.89–0.95) is correctly oriented and stable across the same architectures.

6 Limitations and Future Work

We are explicit about what this paper does not claim. The cost surface is characterized; the investment tiers are mapped; but several important questions remain open.

What we do NOT claim.

- The tensor interface is unfakeable under adversarial training: models might learn to present confident distributions when fabricating.
- Composition preserves epistemic properties: tensor signals may not propagate in recursive LLM pipelines.
- The tensor interface is the optimal design. We demonstrate existence, not optimality.
- These findings generalize to all architectures. Our empirical work spans nine model families up to 235B parameters; closed-weight models whose providers do not expose log-probabilities remain untested.

Future work: Adversarial robustness. Can training find weights that produce confident tensors for fabricated content? Winninger et al. [33] show that mechanistic interpretability tools can identify internal subspaces and craft targeted attacks, confirming that internal signals are rich enough to be both attacked and defended, which is precisely why exporting them matters. The intuition that "you cannot fake the computation you actually performed" may break down under end-to-end adversarial training. Preliminary analysis suggests linguistic and tensor confidence are decoupled in natural outputs, but whether adversarial training can force this coupling remains open. Information-theoretic bounds on signal manipulation are needed.

Future work: Compositional integrity. Does tensor composition preserve epistemic properties in recursive settings? When LLM outputs feed into subsequent LLM inputs, the epistemic trace must either be preserved, regenerated, or discarded. Designing interfaces that maintain epistemic integrity across composition is an open architectural problem.

Cross-architecture generalization. Our bounded verification experiment tests four model families (OLMo-3, Llama-3.1, Qwen3, Mistral) and finds consistent tensor advantage across all of them. Mean pairwise Spearman $\rho = 0.762$ for entropy signals on the same queries supports the interpretation that the signal is architectural rather than model-specific. We validated this finding via API using Together.ai's serverless endpoints across five additional architectures spanning 4B to 235B parameters, including mixture-of-experts models (Llama-4 Maverick 128E, Qwen3-235B) and a Google-family architecture (Gemma 3n). Using only top-5 log-probabilities (renormalized as a lower bound on full-vocabulary entropy), all five models discriminate knowable from unknowable queries with AUC 0.65–0.88 (mean entropy). Preliminary analysis suggests optimal aggregation is architecture-dependent: peak entropy outperforms mean entropy for mixture-of-experts models (AUC $0.65 \rightarrow 0.90$ for Llama-4 Maverick), while entropy variance is most discriminative for dense models. With architecture-appropriate aggregation, all models achieve AUC > 0.85. Cross-model $\rho = 0.36$ in the API set is lower than the local 0.762, likely reflecting the greater architectural diversity (five families, 4B–235B) and the top-5 approximation, rather than signal absence.

Signal access as provider choice. The tensor signals we measure are byproducts of inference that providers can expose or withhold. In practice, access has eroded as models have grown more capable: log-probabilities were available on early completion endpoints but remain unsupported for the newest reasoning models, which is precisely the systems whose outputs are hardest to verify. Responsibility Concentration (Theorem 3.13) formalizes the consequence: when the provider retains exclusive access to epistemic telemetry, only the provider can verify epistemic honesty. Whether to export the tensor interface is a policy decision with direct implications for who bears verification cost.

Model specificity. Our topological analysis (§4) uses OLMo [22], which permits full inspection of intermediate layer representations. The bounded verification experiment (Table 2) tests four architectures spanning 4B–8B parameters and four distinct training pipelines; API validation extends to five additional architectures spanning 4B–235B, including mixture-of-experts models. We do not claim exhaustive coverage, but the consistency across nine model families and two observation regimes (full vocabulary and top-5 log-probabilities) suggests the findings are not idiosyncratic.

Domain-specific calibration. The citation entropy inversion (§5.1) supports the interpretation that the tensor measures generation confidence, not truthfulness directly. This blind spot is precisely where bounded verification is cheapest: citation verification has low compositional complexity (check the DOI, query the database), placing it well within a bounded supervisor's budget. A related confound appears with epistemic refusals: models trained to decline unknowable queries produce low-entropy refusals that are entropically indistinguishable from confident factual answers. As models become more epistemically honest, the entropy signal's ability to detect dishonesty *degrades* because honest refusals and honest facts occupy the same region of entropy space.

Sufficiency gap. The three architectural principles (State Exteriority, Verification Independence, Provenance Binding) are *necessary* conditions for escaping the impossibility. We do not prove they are *sufficient.* A system satisfying all three might still fail epistemic honesty through implementation bugs, adversarial inputs, or unforeseen failure modes. Sufficiency requires additional formal treatment.

The engineering decision. How much epistemic assurance a system needs is an engineering decision, not a research question. A brainstorming tool needs none. A medical summary system needs substantial assurance. A legal research system needs the highest tier available. This paper provides the cost surface for making that decision: what each level of verification investment buys, where the text-channel ceiling lies, what the tensor interface adds, and how content format modulates the tradeoff. The contribution is the map. The territory is the system you are building.

7 Related Work

Our contribution is a formal impossibility result. Prior work documents symptoms; we formalize one structural cause. Prior work proposes treatments; we explain why they cannot fully resolve the verification problem under text-only observation.

Hallucination taxonomies. Extensive empirical work catalogs LLM fabrication patterns: citation hallucination in academic contexts [2], medical misinformation [29], and legal fabrication [9]. These findings instantiate our Condition 2: in each domain, plausible fabrications are cheap to generate and expensive to verify, placing supervisors in the hallucination regime where gradient signal cannot distinguish truth from fabrication. The taxonomies document the symptom; our theorems identify the structural cause.

Model internals. A substantial body of work examines model internal states: transformer circuits [10], sparse features [6], Activation Oracles [16] that train LLMs to interpret other models' internals, and the "Assistant Axis" [20] showing that internal persona state diverges from text behavior. Our work is complementary but narrower: we do not require understanding a model's learned circuits, only that inference-time signals differ measurably between grounded and fabricated generation. Interpretability asks why; observability asks whether.

RLHF critiques. Work on sycophancy [25], reward hacking [12], and the tension between helpfulness and honesty [5] demonstrates that preference optimization can degrade factual accuracy. Our Theorem 3.8 formalizes why: bounded supervisors cannot distinguish plausible fabrications from truthful responses, so the learning signal is systematically corrupted.

Uncertainty quantification. Calibration methods, conformal prediction [3], and verbalized uncertainty [14] attempt to surface model confidence through the text channel. Kuhn et al. [17] advance this with semantic entropy: sampling multiple responses and clustering by meaning. This is the most sophisticated text-channel approach to date, partially addressing gaming by measuring agreement *between* outputs. But it remains text-channel: the clustering operates on generated text, and a bounded supervisor evaluating semantic equivalence remains subject to gaming in the limit. Our tensor interface extracts signals from the computation that produced a single output, which are the signals the model cannot independently control. Semantic entropy and tensor observability are complementary: one measures output-level consistency, the other computation-level confidence.

Impossibility results in LLMs. Xu et al. [34] prove that hallucination is inevitable for LLMs over the class of computable functions, which is a computational limit. Our result is orthogonal: we prove that even if a system could be honest, a bounded supervisor observing only text cannot verify that honesty, which is an observational limit. Their impossibility concerns what LLMs can compute; ours concerns what supervisors can verify. Both are structural, but they constrain different dimensions of system design.

Impossibility results in distributed systems. The Fischer-Lynch-Paterson theorem [11] proved that consensus is impossible under asynchrony with one faulty process. This did not end distributed systems research; it clarified the design space. Paxos [18], Raft [23], and modern consensus protocols work *around* FLP by relaxing assumptions or strengthening guarantees. Our result plays an analogous role: text-only observation models with bounded supervision cannot guarantee epistemic honesty, but systems that export structured epistemic state can escape the impossibility class.

Epistemic evaluation and agent architectures. Wang et al. [30] note that current interfaces lack structured ways to specify uncertainty or sourcing; we provide the formal impossibility explaining *why* text-only evaluation cannot suffice. Chen et al. [8] propose architectures with explicit propositions and justification graphs, which are closest in spirit to our escape conditions, but frame these as *replacements* for stochastic generation. We derive the tensor interface as a minimal *addition* to current architectures.

The gap we fill is formalization: we prove *why* the problem is structural and identify architectural properties that enable escaping it.

8 Conclusion

Systems incorporating language model components must decide how much to spend on verification and where to spend it. This paper provides the cost surface for that decision. Epistemic honesty is unverifiable under text-only observation with bounded supervision, which is a structural property of the interface, not a capability gap. Systems that export structured epistemic state escape this class. Under length-controlled evaluation, no text-channel feature generalizes across architectures; per-token entropy is the only signal that does. Tensor-guided judges outperform text baselines at every budget level (+2.5–3.9pp). The engineering question is not whether to verify, but how much verification to buy and at which tier.

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